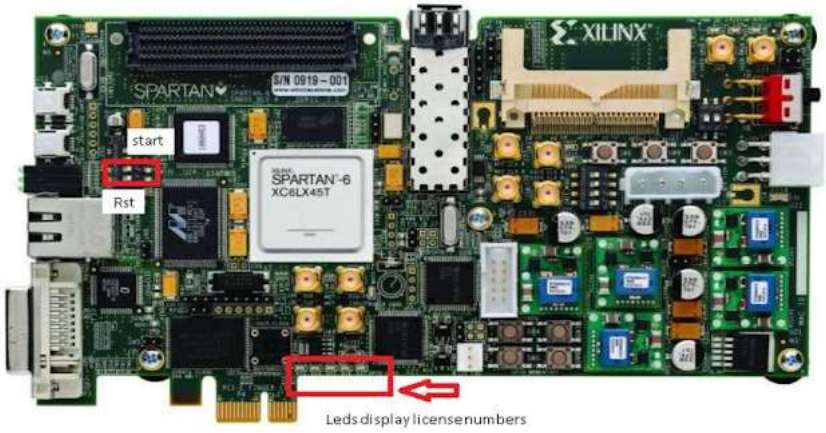
Verilog code for Microcontroller (Part 3- Verilog code)

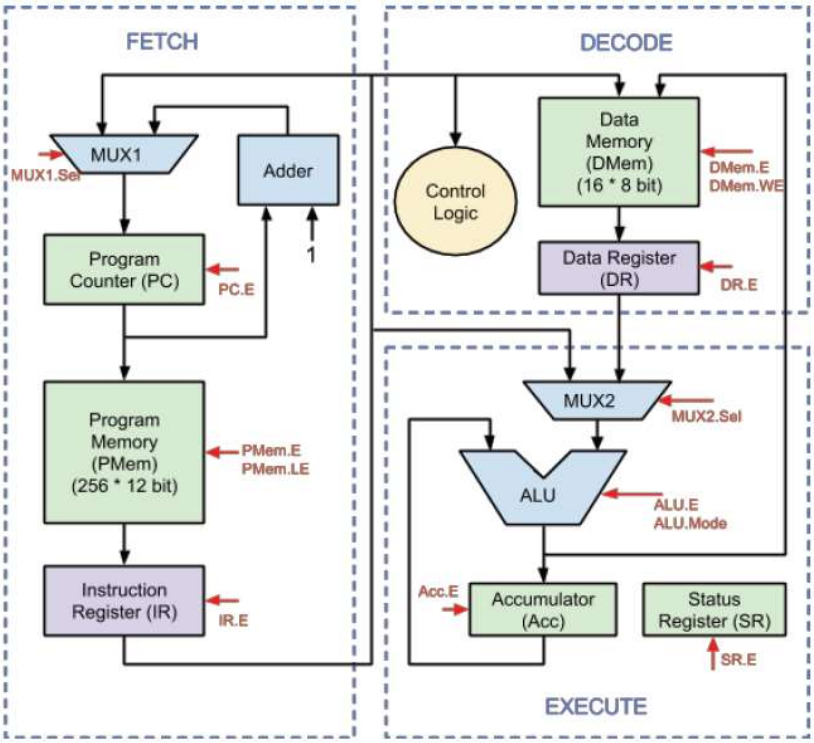
In this project, Verilog code for a microcontroller is presented. The microcontroller architecture and instruction set are shown in part 1 and part 2.



Please visit part-1 and part-2 first before reading code.

Specification of the Microcontroller

Architecture Design for the Microcontroller



First is the Verilog code for ALU unit:

```
// fpga4student.com: FPGA project, Verilog project, VHDL project
// Verilog project: Verilog code for Microcontroller
// Verilog code for ALU
module ALU(  input [7:0] Operand1,Operand2,
  input E,
  input [3:0] Mode,
  input [3:0] CFlags,
  output  [7:0] Out,
  output  [3:0] Flags
// the Z (zero), C (carry), S (sign),O (overflow) bits, from MSB to LSB, connected
```

```
   );
  wire Z,S,O;
  reg CarryOut;
  reg [7:0] Out_ALU;
  always @(*)
  begin
  case(Mode)
  4'b0000: {CarryOut,Out_ALU} = Operand1 + Operand2;
  4'b0001: begin Out_ALU = Operand1 -  Operand2;
  CarryOut = !Out_ALU[7];
  end
  4'b0010: Out_ALU = Operand1;
  4'b0011: Out_ALU = Operand2;
  4'b0100: Out_ALU = Operand1 & Operand2;
  4'b0101: Out_ALU = Operand1 | Operand2;
  4'b0110: Out_ALU = Operand1 ^ Operand2;
  4'b0111: begin
  Out_ALU = Operand2 - Operand1;
  CarryOut = !Out_ALU[7];
  end
  4'b1000: {CarryOut,Out_ALU} = Operand2 + 8'h1;
  4'b1001: begin
  Out_ALU = Operand2 - 8'h1;
  CarryOut = !Out_ALU[7];
  end
  4'b1010: Out_ALU = (Operand2 << Operand1[2:0])| ( Operand2 >> Operand1[2:0]);
  4'b1011: Out_ALU = (Operand2 >> Operand1[2:0])| ( Operand2 << Operand1[2:0]);
  4'b1100: Out_ALU = Operand2 << Operand1[2:0];
  4'b1101: Out_ALU = Operand2 >> Operand1[2:0];
  4'b1110: Out_ALU = Operand2 >>> Operand1[2:0];
  4'b1111: begin
 Out_ALU = 8'h0 - Operand2;
 CarryOut = !Out_ALU[7];
 end
 default: Out_ALU = Operand2;
 endcase
 end
 assign O = Out_ALU[7] ^ Out_ALU[6];
 assign Z = (Out_ALU == 0)? 1'b1 : 1'b0;
 assign S = Out_ALU[7];
 assign Flags = {Z,CarryOut,S,O};
 assign Out = Out_ALU;
// fpga4student.com: FPGA project, Verilog project, VHDL project
endmodule
```

Next is the Verilog code for data memory:

```
// fpga4student.com: FPGA project, Verilog project, VHDL project
// Verilog project: Verilog code for Microcontroller
// Verilog code for data memory
 module DMem( input clk,
     input E, // Enable port
     input WE, // Write enable port
     input [3:0] Addr, // Address port
     input [7:0] DI, // Data input port
     output [7:0] DO // Data output port
    );
reg [7:0] data_mem [255:0];
// fpga4student.com: FPGA project, Verilog project, VHDL project
```

```
always @(posedge clk)
begin
 if(E==1 && WE ==1)
  data_mem[Addr] <= DI;
end
 assign DO = (E ==1 )? data_mem[Addr]:0;
endmodule
```

Verilog code for adder and MUX1:

```
// Verilog project: Verilog code for microcontroller
// Submodule: Verilog code for Adder and MUXs
module adder( input [7:0] In,
      output [7:0] Out
    );

assign Out = In + 1;

endmodule
module MUX1( input [7:0] In1,In2,
     input Sel,
     output [7:0] Out
    );

assign Out = (Sel==1)? In1: In2;
endmodule
```

Continue on the Verilog code for program memory:

```
// Verilog project: Verilog code for microcontroller
// Submodule: Verilog code for Adder and MUXs// fpga4student.com: FPGA project, Ve
// Verilog project: Verilog code for microcontroller
module PMem( input clk,
   input E, // Enable port
  input [7:0] Addr, // Address port
  output [11:0] I, // Instruction port
  // 3 special ports are used to load program to the memory
  input LE, // Load enable port
  input[7:0] LA, // Load address port
  input [11:0] LI//Load instruction port
   );
 reg [11:0] Prog_Mem[255:0] ;
// fpga4student.com: FPGA project, Verilog project, VHDL project
always @(posedge clk)
 begin
 if(LE == 1) begin
 Prog_Mem[LA] <= LI;
 end
 end
 assign I =  (E == 1) ?  Prog_Mem[Addr]: 0 ;
endmodule
module adder( input [7:0] In,
      output [7:0] Out
    );

assign Out = In + 1;

endmodule
module MUX1( input [7:0] In1,In2,
     input Sel,
     output [7:0] Out
    );
```

```
assign Out = (Sel==1)? In1: In2;
endmodule
```

Next, the Verilog code for Control unit:

```
// fpga4student.com: FPGA project, Verilog project, VHDL project
// Verilog project: Verilog code for microcontroller
// Control unit in Verilog
module Control_Logic( input[1:0] stage,
  input [11:0] IR,
  input [3:0] SR,
 output  reg PC_E,Acc_E,SR_E,IR_E,DR_E,PMem_E,DMem_E,DMem_WE,ALU_E,MUX1_Sel,MUX2_S
  output reg [3:0] ALU_Mode
    );
 parameter LOAD = 2'b00,FETCH = 2'b01, DECODE = 2'b10, EXECUTE = 2'b11;
 always @(*)
 begin
 PMem_LE = 0;
 PC_E = 0;
 Acc_E = 0;
 SR_E = 0;
 IR_E = 0;
 DR_E = 0;
 PMem_E = 0;
 DMem_E = 0;
 DMem_WE = 0;
 ALU_E  =0;
 ALU_Mode = 4'd0;
 MUX1_Sel = 0;
 MUX2_Sel = 0;
 if(stage== LOAD )
 begin
 PMem_LE = 1;
 PMem_E = 1;
 end
 else if(stage== FETCH ) begin
 IR_E = 1;
 PMem_E = 1;
 end
 else if(stage== DECODE ) begin
 if( IR[11:9] == 3'b001)
 begin
 DR_E = 1;
 DMem_E = 1;
 end
 else
 begin
 DR_E = 0;
 DMem_E = 0;
 end
 end
 else if(stage== EXECUTE )
 begin
 if(IR[11]==1) begin // ALU I-type
 PC_E = 1;
 Acc_E = 1;
 SR_E = 1;
 ALU_E = 1;
 ALU_Mode = IR[10:8];
 MUX1_Sel = 1;
 MUX2_Sel = 0;
 end
 else if(IR[10]==1) // JZ, JC,JS, JO
 begin
```

```
 PC_E = 1;
 MUX1_Sel = SR[IR[9:8]];
 end
 else if(IR[9]==1)
 begin
 PC_E = 1;
 Acc_E = IR[8];
 SR_E = 1;
 DMem_E = !IR[8];
 DMem_WE = !IR[8];
 ALU_E = 1;
 ALU_Mode = IR[7:4];
 MUX1_Sel = 1;
 MUX2_Sel = 1;
 end
 else if(IR[8]==0)
 begin
 PC_E = 1;
 MUX1_Sel = 1;
 end
 else
 begin
 PC_E = 1;
 MUX1_Sel = 0;
 end
 end
 end
endmodule
```

Finally, the Verilog code for microcontroller:

```
// The microcontroller (similar to Microchip PIC12, simplified extensively, not co
// instruction set) in this project is a 3 cycle nonpipeline
// 8 bit Harvard structure microcontroller, illustrated as the following block dia
// fpga4student.com: FPGA project, Verilog project, VHDL project
// Verilog project: Verilog code for microcontroller
module MicroController(input clk,rst
    );
  parameter LOAD = 2'b00,FETCH = 2'b01, DECODE = 2'b10, EXECUTE = 2'b11;
  reg [1:0] current_state,next_state;
  reg [11:0] program_mem[9:0];
  reg load_done;
  reg[7:0] load_addr;
  wire[11:0] load_instr;
  reg[7:0] PC,  DR, Acc;
  reg [11:0] IR;
  reg [3:0]  SR;
  wire PC_E,Acc_E,SR_E,DR_E,IR_E;
 reg PC_clr,Acc_clr,SR_clr,DR_clr,IR_clr;
 wire [7:0] PC_updated,DR_updated;
 wire[11:0] IR_updated;
 wire[3:0] SR_updated;
 wire PMem_E,DMem_E,DMem_WE,ALU_E,PMem_LE,MUX1_Sel,MUX2_Sel;
 wire [3:0] ALU_Mode;
 wire [7:0] Adder_Out;
 wire [7:0] ALU_Out,ALU_Oper2;
 // LOAD instruction memory
 initial
 begin
 $readmemb("program_mem.dat", program_mem,0,9);
 end
 // ALU
 ALU ALU_unit( .Operand1(Acc),
      .Operand2(ALU_Oper2),
```

```
    .E(ALU_E),
    .Mode(ALU_Mode),
    .CFlags(SR),
       .Out(ALU_Out),
       .Flags(SR_updated) // the Z (zero), C (carry), S (sign),O (overflow) bits, f
  );
 // MUX2
 MUX1 MUX2_unit( .In2(IR[7:0]),.In1(DR),
  .Sel(MUX2_Sel),
  .Out(ALU_Oper2)
    );
 // Data Memory
 DMem DMem_unit( .clk(clk),
  .E(DMem_E), // Enable port
  .WE(DMem_WE), // Write enable port
  .Addr(IR[3:0]), // Address port
  .DI(ALU_Out), // Data input port
  .DO(DR_updated) // Data output port
    );
 // Program memory
 PMem PMem_unit( .clk(clk),
  .E(PMem_E), // Enable port
  .Addr(PC), // Address port
  .I(IR_updated), // Instruction port
  // 3 special ports are used to load program to the memory
  .LE(PMem_LE), // Load enable port
  .LA(load_addr), // Load address port
  .LI(load_instr)//Load instruction port
   );
 // PC ADder
 adder PC_Adder_unit( .In(PC),
   .Out(Adder_Out)
    );
// fpga4student.com: FPGA project, Verilog project, VHDL project
// MUX1
 MUX1 MUX1_unit( .In2(IR[7:0]),.In1(Adder_Out),
  .Sel(MUX1_Sel),
  .Out(PC_updated)
    );
 // Control logic
 Control_Logic Control_Logic_Unit( .stage(current_state),
  .IR(IR),
  .SR(SR),
  .PC_E(PC_E),
  .Acc_E(Acc_E),
  .SR_E(SR_E),
  .IR_E(IR_E),
  .DR_E(DR_E),
  .PMem_E(PMem_E),
  .DMem_E(DMem_E),
  .DMem_WE(DMem_WE),
  .ALU_E(ALU_E),
  .MUX1_Sel(MUX1_Sel),
  .MUX2_Sel(MUX2_Sel),
  .PMem_LE(PMem_LE),
  .ALU_Mode(ALU_Mode)
    );
// fpga4student.com: FPGA project, Verilog project, VHDL project
 // LOAD
 always @(posedge clk)
 begin
 if(rst==1) begin
 load_addr <= 0;
```

```
 load_done <= 1'b0;
 end
 else if(PMem_LE==1)
 begin
 load_addr <= load_addr + 8'd1;
 if(load_addr == 8'd9)
 begin
 load_addr <= 8'd0;
 load_done <= 1'b1;
 end
 else
 begin
 load_done <= 1'b0;
 end
 end
 end
// fpga4student.com: FPGA project, Verilog project, VHDL project
 assign load_instr = program_mem[load_addr];
 // next state
 always @(posedge clk)
 begin
 if(rst==1)
 current_state <= LOAD;
 else
 current_state <= next_state;
 end
 always @(*)
 begin
 PC_clr = 0;
 Acc_clr = 0;
 SR_clr = 0;
 DR_clr = 0;
 IR_clr = 0;
 case(current_state)
 LOAD: begin
 if(load_done==1) begin
 next_state = FETCH;
 PC_clr = 1;
 Acc_clr = 1;
 SR_clr = 1;
 DR_clr = 1;
 IR_clr = 1;
 end
 else
 next_state = LOAD;
 end
 FETCH: begin
 next_state = DECODE;
 end
 DECODE: begin
 next_state = EXECUTE;
 end
 EXECUTE: begin
 next_state = FETCH;
 end
 endcase
 end
// fpga4student.com: FPGA project, Verilog project, VHDL project
 // 3 programmer visible register
 always @(posedge clk)
 begin
 if(rst==1)
 begin
```

```
 PC <= 8'd0;
 Acc <= 8'd0;
 SR <= 4'd0;
 end
 else
 begin
 if(PC_E==1'd1)
 PC <= PC_updated;
 else if (PC_clr==1)
 PC <= 8'd0;
 if(Acc_E==1'd1)
 Acc <= ALU_Out;
 else if (Acc_clr==1)
 Acc <= 8'd0;
 if(SR_E==1'd1)
 SR <= SR_updated;
 else if (SR_clr==1)
 SR <= 4'd0;
 end
 end
// fpga4student.com: FPGA project, Verilog project, VHDL project
 // 2 programmer invisible register
 always @(posedge clk)
 begin
 if(DR_E==1'd1)
 DR <= DR_updated;
 else if (DR_clr==1)
 DR  <= 8'd0;
 if(IR_E==1'd1)
 IR <= IR_updated;
 else if(IR_clr==1)
 IR <= 12'd0;
 end
endmodule
```

Verilog testbench code to verify the microcontroller:

```
module MCU_tb;
// fpga4student.com: FPGA project, Verilog project, VHDL project
// Verilog project: Verilog code for microcontroller
// Inputs
reg clk;
reg rst;
// Instantiate the Unit Under Test (UUT)
MicroController uut (
.clk(clk),
.rst(rst)
);
initial begin
// Initialize Inputs
rst = 1;
// Wait 100 ns for global reset to finish
#100;
rst = 0;
end
initial begin
clk = 0;
forever #10 clk = ~clk;
end
endmodule
```

Test program 1:

```
0000_0000_0000
1011_0000_0001
0010_0010_0000
1011_0000_0000
0011_0011_0000
0001_0000_0101
0000_0000_0000
0000_0000_0000
0000_0000_0000
0000_0000_0000
```

Test program 2:

```
0000_0000_0000
1011_0000_0001
0010_0010_0000
0011_0000_0000
0010_0000_0000
0011_0001_0000
0010_0001_0000
0011_0111_0000
0010_0111_0000
0001_0000_1001
```

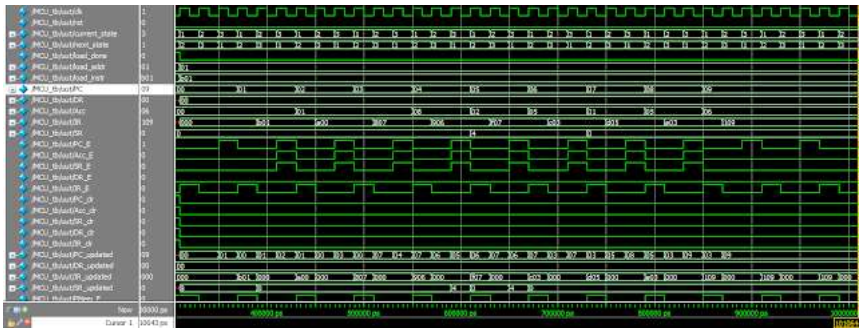
Test program 3:

```
0000 0000 0000
1011_0000_0101
0010_0010_0000
0010_0010_0001
0010 0010 0010
1011_0000_0011
0010_0100_0000
0010_0101_0001
0010 0110 0010
0001_0000_1001
```

Test program 4:

```
0000_0000_0000
1011_0000_0101
1010_0000_0000
1000_0000_0111
1001_0000_0110
1111_0000_0111
1100_0000_0011
1101_0000_0101
1110_0000_0011
0001_0000_1001
```

Use the four test program to run simulation and check if the microcontroller is working correctly.



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